Chapter 2

Futures Markets, Clearing, and Central Counterparties

A forward contract is a promise. Its value depends entirely on the promisor being willing and able to perform, years after the promise was made, at a price that may by then look catastrophic. Futures markets exist to remove that dependence, and the machinery they use to do so—standardization, daily settlement, and a central counterparty standing between every buyer and seller—is the subject of this chapter.

That machinery works well enough that it is easy to take for granted. This chapter deliberately does not. The events of 2022 turned questions that had been footnotes in earlier textbooks into first-order concerns: how much margin a clearing house should demand, what happens when it demands too much too fast, and who absorbs the loss if a large member fails. Sections 2.8 through 2.10 take them up directly.

Sections 2.1 and 2.2 are the contract: what a futures specification fixes and why the futures price converges to spot. Sections 2.3 to 2.6 are the machinery: margin and daily settlement, the central counterparty, clearing members and their clients, and delivery. Section 2.7 covers regulation and manipulation; Sections 2.8 to 2.10 are the three questions above; Section 2.11 is the OTC markets, cleared and bilateral; Section 2.12 is who trades and how orders reach the market; and Section 2.13 closes by setting forwards against futures.

2.1 FUTURES CONTRACTS AND THEIR SPECIFICATION

A **futures contract** is an agreement to buy or sell an asset at a specified future time for a price agreed today. What distinguishes it from a forward is that it is standardized and exchange-traded, which permits a liquid secondary market: a position can be closed by taking the opposite position rather than by negotiating with the original counterparty.

Standardization means the exchange, not the parties, specifies everything except the price:

- **The asset.** For commodities, a grade and a set of permitted substitutions with price adjustments. This is not a detail, since a contract's deliverable grade determines what it actually tracks.
- **The contract size.** Too small and commissions dominate; too large and smaller hedgers are excluded. Exchanges sometimes list "mini" contracts for this reason.
- **Delivery arrangements.** Where delivery may be made, and for financial contracts whether delivery occurs at all or the contract is cash-settled against a published index.
- **Delivery months and the last trading day.**
- **Price quotation, tick size (the smallest permitted price change), and on some contracts daily price limits and position limits.**

Table 2.1 shows the specification of one such contract.

Table 2.1: The light sweet crude oil futures contract. Terms as of press date.

Exchange	New York Mercantile Exchange
Ticker	CL
Contract size	1,000 barrels
Deliverable grade	West Texas Intermediate light sweet crude, with listed substitutable grades
Delivery location	Cushing, Oklahoma
Delivery months	Every month, several years forward
Price quotation	US dollars and cents per barrel
Tick size	$0.01 per barrel, or $10 per contract
Last trading day	Third business day before the 25th of the preceding month
Position limits	Set by the exchange, tightening near delivery

Two entries repay attention. The *deliverable grade* determines what the contract tracks: a producer whose output differs from the deliverable grade is exposed to the difference, which is the basis risk of Chapter 3. And the *delivery location* matters as much as the grade: a contract deliverable at an inland hub prices differently from one deliverable at a port, because the two are connected only by pipeline capacity that can bind.

Table 2.2 shows that crude oil contract once it is trading.

Table 2.2: Quotes by delivery month. Illustrative, not market data.

Delivery	Settle ($/bbl)	Change	Volume	Open interest
December	78.29	+0.24	412,880	128,450
January	78.59	+0.21	198,340	265,910
February	78.88	+0.18	76,220	142,300
March	79.18	+0.16	41,150	98,660
June	80.07	+0.11	12,480	54,120
December (next)	81.90	+0.07	6,310	71,940

Five things can be read off it, and most are taken up again later:

- *The settle column is a forward curve*: the price of the same asset for delivery at each of six dates, generated here from Chapter 5's cost-of-carry relation at a spot price of $78.00 a barrel and a net carry rate of 4.5% a year, continuously compounded, with the six contracts delivering 1, 2, 3, 4, 7 and 13 months out. It is the object Section 2.12 says these markets exist to produce. It slopes upward here because storage and financing costs exceed any convenience yield (the benefit of holding the physical commodity rather than a claim on it), which is the storable-commodity case of Section 2.2.
- *Volume and open interest are different quantities and peak in different months*: December trades most because positions are being rolled out of it (closed in the expiring month and reopened in the next), while January holds the most open interest because that is where they are going.
- *Liquidity thins with maturity*, which is why the stack-and-roll hedge of Chapter 3 exists at all: a hedger wanting ten-year protection cannot simply buy it in the ten-year contract, so it holds the whole exposure in the liquid near contract and rolls it forward as each one expires.
- The December contract a year out carries more open interest than the nearer June, because annual hedging programs settle on December dates.
- The tick from Table 2.1 converts the change column into money: December's $0.24 is $240 a contract.

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Price limits and position limits. A **daily price limit** halts trading if the price moves more than a specified amount from the previous settlement. The intention is to prevent large moves driven by disorderly trading. The effect is contested: a limit does not remove the information causing the move, and a market that is limit-locked is one in which no one can trade at all, which converts a price problem into a liquidity problem. **Position limits** cap the contracts a speculator may hold, to reduce the scope for cornering the market, that is, holding a position large enough that the shorts must buy back from the party that also controls the deliverable supply, which Section 2.7 takes up. They typically tighten as delivery approaches, when the deliverable supply is most easily squeezed.

Volume and open interest

Two quantities are reported for every contract and are constantly confused. **Volume** is the number of contracts traded during the day. **Open interest** is the number of contracts outstanding at its end, counted on one side only, since every long is matched by a short.

They differ because a trade may open positions, close them, or transfer one from one holder to another. Only the first and second change open interest.

Example 2.1 Volume and open interest

Trace a newly listed contract over four days.

Day	Trade	Volume	Change in OI	Open interest
1	A opens long 10; B opens short 10	10	+10	10
2	C opens long 5; D opens short 5	5	+5	15
3	A closes 4; C opens 4	4	0	15
4	A closes 6; B closes 6	6	−6	9

On day 3 one participant left and another arrived, so the number of outstanding contracts was unchanged: the position was transferred, not created or extinguished. On day 4 both sides closed, so six contracts ceased to exist.

The distinction carries information. Rising prices on rising open interest suggest new money entering on the long side; rising prices on *falling* open interest suggest shorts closing out, which is a different phenomenon and typically less persistent.

2.2 CONVERGENCE TO THE SPOT PRICE

As delivery approaches, the futures price converges to the spot price. The argument is a simple arbitrage.

Suppose during the delivery period the futures price is above the spot price. A trader can sell the futures, buy the asset, and deliver immediately, earning the difference. This is riskless and will be done until the gap closes. If instead the futures price is below the spot price, those who wish to acquire the asset will buy futures and wait for delivery, pushing the futures price up.

Neither argument requires any view about where the price is going. It follows that at expiration the two prices must be essentially equal, and the futures contract can be treated as a claim on the spot asset at that date.

The futures price may approach the spot price from above (the normal pattern for a storable commodity, where the futures price includes storage and financing costs that shrink as delivery nears) or from below, which happens when holding the physical asset conveys a benefit that the futures contract does not. Chapter 5 develops what determines the gap *before* expiration; the point here is only that it must close.

What converges is the gap, not the prices. Spot and futures prices both move throughout,

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and neither approaches delivery along any predictable line. The gap between them (the spot price minus the futures price, called the basis) *narrows to zero anyway*, because the arbitrage that forces it does not care about the route.

The convergence is what makes a futures contract useful as a hedge at all. A hedger who holds the physical asset and shorts futures relies on the two prices ending up together; if they did not, the hedge would leave a residual whose size was unknown in advance. Chapter 3 examines what happens when the hedge is lifted *before* delivery, when convergence is incomplete and the residual is exactly the basis risk. Figure 3.1 draws the convergence with that residual marked.

2.3 MARGIN AND DAILY SETTLEMENT

The exchange's central problem is that a futures contract can accumulate large losses for one side over its life, and the losing party may prefer to walk away. The solution is not to check creditworthiness but to make the question irrelevant: **losses are collected daily, in cash, before they can grow**.

The margin account. On opening a position, a trader deposits **initial margin** with the broker. At the end of each trading day the account is adjusted for that day's gain or loss, called **marking to market**, or **daily settlement**. Gains are credited to the account, and any balance above the initial margin may be withdrawn; losses are debited from it.

The balance need not stay at the initial margin. It may fall as far as a lower threshold, the **maintenance margin**, before anything happens; only if it falls below that does the trader receive a **margin call**, and must then restore the balance to the *initial* margin level. If the call is not paid, the broker closes the position. The daily gains and losses themselves, settled in cash, are the **variation margin**; older texts give that name to the deposit that answers a call, and Example 2.2 shows why the two are better kept apart.

What margin is posted in. Initial margin need not be cash. Clearing houses also accept Treasury bills and other government securities, valued after a haircut (a deduction from their market value; Section 4.1), and many hedgers post them for exactly that reason: a bill keeps earning its yield while it sits in the account. Cash may earn interest too, but at a rate the broker or clearing house sets, usually below the overnight benchmark, so on a large position the forgone interest is a real cost of hedging. Variation margin is different. It is settled in cash every day, and the interest earned on gains or paid to finance losses is what can make a futures price differ from a forward price (Section 5.12).

Example 2.2 Operation of a margin account

An investor buys two gold futures contracts, each for 100 ounces, at a futures price of $3,200 per ounce, a position of 200 ounces and $640,000 of notional. The clearing house sets maintenance margin at $15,000 per contract and initial margin ten percent above it at $16,500, so the investor posts $33,000 and must keep the account above $30,000. The initial requirement is about 5% of notional, the order of magnitude a liquid futures contract attracts, and it exceeds the maintenance level by a set proportion.

Day	Futures price	Daily gain	Cumulative gain	Balance	Margin call
0	3,200.00	—	—	33,000	—
1	3,188.00	−2,400	−2,400	30,600	—
2	3,176.00	−2,400	−4,800	28,200	4,800
3	3,168.00	−1,600	−6,400	31,400	—
4	3,190.00	+4,400	−2,000	35,800	—

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On day 2 the balance of \$28,200 falls below the \$30,000 maintenance level, so the investor must deposit \$4,800 to restore the \$33,000 initial level. Day 3 therefore starts from \$33,000 and closes at \$31,400, still above maintenance, so no further call; on day 4 the price recovers to leave \$35,800.

Check the arithmetic overall: the price fell \$10 from 3,200 to 3,190, a cumulative loss of $10 \times 200 = \$2{,}000$. The investor deposited $33{,}000 + 4{,}800 = \$37{,}800$, and $37{,}800 - 2{,}000 = \$35{,}800$.

The daily gain column is the variation margin. Each entry is the day's mark-to-market, debited from the account when the position loses and credited when it gains, and it is paid every day whether or not any call is made. The \$4,800 on day 2 is not variation margin but a top-up of *initial* margin. The two are indistinguishable in this table because a futures account runs them through one balance; Section 8.2 separates them. The separation matters most to a hedger: variation margin on the futures leg must be paid in cash every day, even while the position it hedges gains as much on paper, so a hedge that is working economically can still run its owner out of cash.

Two further features deserve attention. The investor was called for cash on day 2 even though the position was still open and might recover, as it did. And the \$2,800 standing above the initial requirement on day 4 could be withdrawn immediately. Futures positions consume and release cash continuously, which is a different property from owning the underlying asset.

What the daily settlements add up to. The example's cross-check is an instance of a general fact. A position opened at F_1 and closed at F_2, settled every day in between, collects

$$F_2 - F_1 \text{ per unit if long}, \qquad F_1 - F_2 \text{ if short}, \tag{2.1}$$

whatever route the price took in between. Each intermediate price enters two consecutive settlements with opposite signs and cancels, so the sum telescopes to the difference between the first and last. Equation (2.1) is what allows a futures position to be treated as a single payoff at the date it is closed, which is how Chapter 3 reads a hedge and how Chapter 5 prices one. What does *not* cancel is the timing: the same total arrives as a stream rather than a lump, which is the subject of Section 2.9 and of Chapter 8, and the reason a futures price and a forward price are not quite the same number.

Why this matters beyond the mechanics. Example 2.2 looks like bookkeeping, and for a two-contract position it is. Scaled up, it is the mechanism behind several of the episodes in this book. A hedger whose futures leg loses money must find cash *today*, while the offsetting gain on the hedged item may not be realizable for years. The hedge is economically sound and the firm can still fail. Chapter 3 describes this happening to Metallgesellschaft; Section 2.9 describes it happening across an entire market in 2022.

And what it does to a speculator. Read the other way, the same table is the risk in a futures position taken for a view. Example 2.2 commits \$33,000 against \$640,000 of exposure, so gold need fall only 5.2% to consume the deposit, and past it the trader owes the rest of the move in cash the next morning; the option of Example 1.5, whose premium bounds the loss, differs in kind. The consequence is not that leverage is large but that it is *continuous*: a view expressed in futures must be solvent every day until it is vindicated, because a call unmet is a position closed by the broker at whatever price the market offers that morning. Section 40.4 is this arithmetic at institutional scale, and Chapter 38 explains why a risk measure built on the distribution of outcomes does not see it: the loss that matters is on the path, not at the horizon.

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How initial margin is set

Example 2.2 took the margin requirement as given. How it is arrived at determines a great deal of what follows in this chapter.

Initial margin is intended to cover the loss a position could suffer over the time it would take to close it out if the holder defaulted (one or two days for a liquid futures contract, five or more for a cleared swap portfolio) at a high confidence level, typically 99% to 99.7%. Clearing houses estimate the distribution of moves from historical data, often by simulating the portfolio's value under several years of observed daily moves and reading off the appropriate quantile. Strictly, this sets what a clearing member posts to the CCP. Exchange rules turn that figure into the customer's maintenance margin, with initial margin set above it (by ten percent in Example 2.2), and a broker may require more of its own clients.

These models have names, and a practitioner meets them before meeting any of the theory. The **Standard Portfolio Analysis of Risk** (SPAN) has been the long-standing framework for listed futures and options: it revalues the portfolio under a fixed grid of price and volatility scenarios and charges the worst outcome, which is why a margin requirement can be reproduced exactly from a published parameter file. Cleared swaps are margined instead by simulation across several years of historical moves, and the listed markets are migrating the same way. *The simulation-based models are the historical simulation of Section 38.5 under another name*, and the Standard Initial Margin Model (SIMM) of Chapter 8 is the bilateral cousin: the same problem, solved by an industry committee rather than by a clearing house.

The feature with the largest practical consequence is that margin is computed on the *portfolio*, not contract by contract. Positions that offset receive credit for it, as Example 2.3 shows.

Example 2.3 Portfolio margining and spread positions

A trader holds 100 long contracts in the March delivery month and 100 short in June. Taken separately, each leg might require $2,000 per contract:

$$200 \text{ contracts} \times \$2{,}000 = \$400{,}000$$

But the two legs nearly offset: a parallel move in the underlying gains on one and loses on the other, and what remains is exposure to the *spread* between the two months, which is far less volatile. The clearing house might set the requirement for the paired position at $200 per contract pair:

$$100 \text{ pairs} \times \$200 = \$20{,}000$$

a twentieth of the gross figure.

This is why calendar spreads, basis trades, and relative-value strategies generally can be run at leverage that would be impossible outright. Chapter 6 makes the point about the cash–futures basis trade, where a hedged position financed by repo (Section 4.1), with the lender withholding only a small haircut of the bond's value, runs at close to fifty times leverage.

It also identifies the danger. The offset is granted on the assumption that the two legs move together. When they stop doing so (when the basis itself dislocates, as in March 2020), the margin credit is withdrawn at the same moment the position starts losing money. This is a more violent version of the procyclicality of Section 2.9.

2.4 CENTRAL COUNTERPARTIES

Daily settlement limits how large a loss can grow. It does not answer who bears the loss if a party fails to pay. That is the role of the **central counterparty**, or clearing house.

When a trade is cleared, the original contract between buyer and seller is replaced, or **novated**, by two contracts, each facing the CCP. The CCP becomes the buyer to every seller and the seller to every buyer. Neither original party has any further exposure to the other; both

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are exposed to the CCP.

Two things the rest of this chapter takes for granted follow from that substitution, and neither would exist without it. The first is **offset**. Section 2.6 observes that most positions are closed before delivery, and Example 2.1 watches open interest fall as traders do it, but closing out is only possible because the contract is with the CCP rather than with whoever sold it. A trader who is long sells to *anyone*, the CCP nets the two positions against each other, and the position is gone. Without novation the trader would hold two live contracts with two different parties, remain exposed to both, and be able to exit only by persuading the original counterparty to tear the trade up.

The second is that the order book can be anonymous. Section 2.1 credits standardization with a liquid secondary market, and the reason is that any two participants wanting the same contract meet in the same place. That works only if neither has to price the other's credit before trading, and novation is what makes the contracts fungible: a March contract is one instrument whatever the identity of the seller. *The CCP is therefore not an adjunct to the futures market but a precondition for it*: liquidity, offset, and the daily settlement of Section 2.3 all rest on it.

The change in structure, from the web in the left panel of Figure 1.2 to the hub in the middle one, is not cosmetic. With n participants trading bilaterally there are up to $n(n-1)/2$ exposures (6 for the four participants drawn there, 15 for six, and tens of thousands for a real market). Clearing replaces them with n. Three benefits follow.

Multilateral netting. A member long with one counterparty and short with another has no net position, and after novation no net exposure. Bilaterally it would have had two.

Uniform, transparent margin. The CCP sets margin by rule rather than by bilateral negotiation, which removes the tendency for the strongest firms to demand collateral from the weakest while extending credit to each other.

Orderly default management. A CCP has a rehearsed procedure for closing out a failed member's positions, typically by auctioning the portfolio to survivors. In a bilateral market each of a failed firm's hundreds of counterparties scrambles simultaneously to replace its own hedge, which is what made September 2008 as disorderly as it was.

The netting benefit is the largest of the three and is worth computing rather than asserting, because its size is what justifies the concentration of risk that comes with it, and because the same arithmetic shows how easily it is given away.

Example 2.4 What multilateral netting is worth, and what fragmentation costs

Five dealers, with every pair's obligations already netted bilaterally, so these ten figures are what a bilateral market leaves outstanding:

Owes		Owes	
A → B	120	C → A	80
A → D	45	C → D	90
B → C	60	D → B	110
B → E	25	D → E	70
E → A	30	E → C	55

Gross bilateral exposure is the sum of these, **685**, because each is owed to a different party and none offsets another.

Novate all ten to one CCP and each dealer holds a single net position:

	A	B	C	D	E
Net position	−55	+145	−55	−45	+10

The five sum to zero, as they must. Total exposure is now the sum of the positive positions,

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155, 22.6% of the bilateral figure, a 77.4% reduction, and no netting agreement, collateral, or credit assessment produced it. It came from replacing ten counterparty relationships with five.

Now split the market. Suppose A, B, and C clear at one CCP, D and E at another, and the six trades that cross the boundary stay bilateral. Exposure rises to **485**, 70.8% of the bilateral figure, giving back 330 of the 530 that netting had saved. Fragmenting a cleared market across two CCPs destroys nearly two-thirds of the benefit of clearing it, and nothing either CCP does (margin, default fund, or waterfall) can win it back, because the lost offsets run between the two.

That arithmetic is the argument in every dispute about where a CCP should sit and about interoperability (links between CCPs that let members of one clear trades against members of another), and it explains why dealers resist being required to clear the same product in two places. The case for one CCP is quantitatively strong, and it is the same calculation that makes that CCP systemically important.

What clearing does not do. It does not eliminate risk; it relocates and concentrates it. Every participant now depends on one institution. That institution's resilience is therefore a matter of public interest, which is what the rest of this chapter is about.

2.5 CLEARING MEMBERS, CLIENTS, AND PORTING

Figure 1.2 implies that everyone faces the CCP directly. Almost no one does. Access is two-tier: a limited number of well-capitalized firms are **clearing members** of the CCP, and everyone else (asset managers, pension funds, corporates, hedge funds) clears *through* a member as a **client**. A client's positions reach the CCP only through its member, so if the member fails, what the client recovers turns entirely on how those positions and their collateral were held.

Segregation. Client collateral must be kept separate from the member's own. Under **individual segregation** each client's collateral sits in its own account; under **omnibus segregation** clients are pooled, which is cheaper and offers less protection, since a shortfall caused by one client in the pool can reach the others. Many clients do not know which they have.

What nets against what. A member holds more than one position at the CCP. Its own trades sit in a **house account**, where Example 2.4's netting applies in full; its clients' sit in separate accounts, and the two never net, which is what segregation means. Whether clients net against one another depends on how the account is margined: margined **gross**, the CCP computes initial margin client by client and adds the results; in a **net** omnibus account the pool offsets first, which is why it is cheaper. *Variation margin nets in every regime*: it settles a loss already suffered, so offsetting it gives nothing away. The choice governs initial margin alone.

Porting. If a clearing member fails, the CCP will attempt to **port** its clients (transfer their positions and collateral to a surviving member) rather than close them out. Porting preserves the client's hedge; closing out destroys it and leaves the client to rebuild in a market that is, by construction, disorderly. *It is also what gross margining buys*: collateral attributable to a client is collateral that can move with them, whereas a netted pool has a single margin figure out of which no one client can be lifted.

But porting depends on a surviving member being willing to accept the client, and that willingness is scarcest exactly when it is needed. A member contemplating a failed competitor's client book during a crisis is asked to take on unfamiliar positions and consume its own capital at the worst moment. Clients therefore hold a protection that is real but conditional—and its condition correlates with the event it protects against.

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The two-tier structure also concentrates the market further than Figure 1.2 suggests: the number of firms willing to offer client clearing has fallen steadily, since the business consumes balance sheet under the leverage ratio (Section 1.12), while earning thin fees. A system designed to reduce concentration risk rests on a shrinking number of intermediaries.

2.6 DELIVERY

Most futures positions are closed before delivery (for financial contracts, the overwhelming majority). But the possibility of delivery is what ties the futures price to the spot price, so the mechanics matter even to those who never use them.

For a physically delivered contract the process runs from the short side. During the delivery month the short may issue a **notice of intention to deliver** to the exchange. The exchange assigns the notice to a long (typically the one with the oldest outstanding position), who must accept delivery and pay the invoice amount. The short chooses *when* within the delivery period to issue notice, and for some contracts *what* to deliver from a permitted set. Where that set spans grades or delivery points, the invoice amount is not the futures price alone: published **price adjustments** add a premium or discount for what was actually delivered, measured against the par grade and location, the ones the quoted futures price refers to. Without them the short would simply deliver the cheapest admissible item and keep the difference. Chapter 6 shows what those choices are worth in the Treasury market, where the adjustment is the conversion factor.

For a physical commodity, everything it costs to hold the asset falls on the short until delivery. Storing it, insuring it, and moving it to an approved delivery point are the short's expense, which is why those costs appear in the forward price of Section 5.8 rather than anywhere in the contract's terms. What happens at delivery then depends on whether the exchange licenses somewhere to put the asset. For metals and grains it does, and delivery is the transfer of a **warehouse receipt** (title to a specified lot already sitting in a licensed warehouse) rather than the movement of anything: storage comes with the title, and the long inherits the charges from that day. Getting the metal or grain physically out is a separate operation with a queue of its own, and Section 36.12 shows that the length of that queue moves the price.

Where no such warehouse stands behind the contract, the obligation is harder than it looks, and it belongs to the long. Crude oil is delivered into tankage at Cushing, Oklahoma, and a long that has not arranged capacity before expiration must find it in a market where everyone else needs it at the same moment. Section 5.9 describes the April 2020 expiration at which there was none to be had and the price settled below zero. *This is the reason most participants close out rather than deliver*, and the reason a retail product that mechanically holds the front contract is not the safe instrument it appears to be.

Financial contracts are usually **cash-settled** instead, against a published settlement index. There is no deliverable asset for a stock index or an overnight interest rate, and cash settlement avoids the artificiality of constructing one. It also removes the squeeze risk that delivery obligations create, at the cost of making the contract dependent on the integrity of the settlement index, which is its own vulnerability, as Chapter 4 shows for LIBOR.

2.7 REGULATION AND MARKET MANIPULATION

Futures markets are regulated to ensure prices are set by genuine supply and demand rather than by participants able to distort them. The characteristic abuse is the **corner** or **squeeze**: a trader accumulates a dominant long futures position *and* control of the deliverable supply, so that shorts approaching delivery must buy back from the same party that controls the physical market. The price rises not because the commodity has become scarce in any economically meaningful sense but because the shorts cannot obtain it.

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The defenses are position limits, tightening into the delivery period; surveillance of large positions; and the exchange's power to intervene by raising margin, imposing liquidation-only trading (positions may be closed but not opened), or in extremis altering delivery terms.

The nickel episode of Section 2.9 is instructive precisely because it was not a manipulation. The large short was a genuine producer hedge, and the price spike was driven by shorts scrambling to cover. Yet the market effect was much the same: prices dislocated from any plausible physical value, participants unable to transact. Regulation aimed at intent does not necessarily protect against outcomes that arise without it.

2.8 THE DEFAULT WATERFALL

A CCP that guarantees performance must be able to absorb the failure of a member. The resources it can call on are arranged in a strict sequence, the **default waterfall**, shown in Figure 2.1. Beyond the defaulter's own margin, the main resource is the **default fund**: a pool to which every clearing member contributes, held to cover losses larger than a defaulter's margin can absorb.

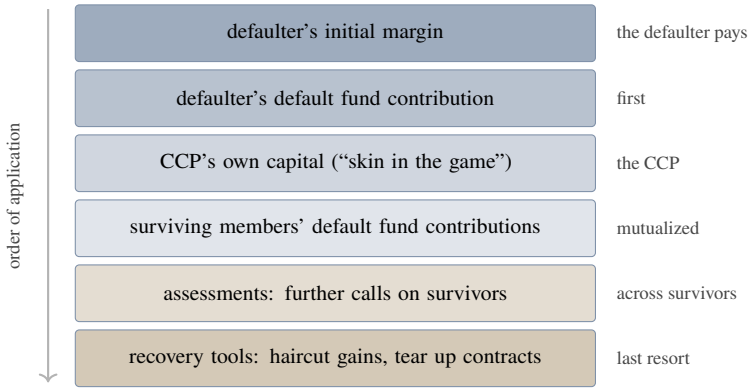


Figure 2.1: The sequence in which a CCP applies resources after a member default.

The ordering embodies a principle: *the defaulter pays first*. Only when the defaulter's own resources are exhausted does anyone else contribute, and the CCP's own capital (its *skin in the game*) sits ahead of surviving members' mutualized funds, so that a CCP which sets margin too low bears the first consequence itself. That layer is small relative to the fund below it, and how small it should be is one of the live disputes in the field. Example 2.5 takes one default through the layers.

Example 2.5 Applying the waterfall

A clearing member defaults. Closing out its portfolio in the market realizes a loss of $900 million. The CCP's resources are:

Layer	Amount ($m)
Defaulter's initial margin	500
Defaulter's default fund contribution	80
CCP skin in the game	50
Surviving members' default fund	2,000

Applying the waterfall in order: the defaulter's margin absorbs $500m, leaving $400m. Its default fund contribution absorbs $80m, leaving $320m. The CCP's own capital absorbs $50m,

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leaving \$270m. That remainder is mutualized across the surviving members, consuming

$$\frac{270}{2{,}000} = 13.5\%$$

of their pooled contributions.

The survivors did nothing wrong, had no exposure to the defaulter bilaterally, and have lost 13.5% of the funds they posted, at a moment when a major member has just failed and markets are presumably under stress. Clearing did not make the loss disappear; it distributed it. That is what "mutualization" means, and members' willingness to accept it is what the system rests on.

2.9 MARGIN PROCYCLICALITY

Initial margin is calibrated to cover a large adverse move over the time it would take to close out a defaulted portfolio, conventionally a high confidence level over a few days. Since estimates of how far prices move are based on recent volatility, **margin requirements rise when volatility rises**. Because volatility is highest in a falling, stressed market, margin moves with the market cycle (lowest in calm, highest in a crisis), and a requirement that moves with the cycle reinforces it. That property is **procyclicality**.

This is prudent for the CCP taken alone and destabilizing for the system taken together. Consider what it means. Volatility spikes; every member's margin requirement increases at once; all of them must find cash simultaneously; and the natural way to raise cash is to sell assets into a market that is already falling. Margin models designed to protect the clearing house can amplify the very move they are protecting against. Example 2.6 puts a size on the jump.

Example 2.6 How fast margin can move

A fund holds a portfolio whose initial margin is set to cover a three-standard-deviation adverse move over three days, about 99.9% one-sided. Under normal conditions, with annualized volatility of 15%, the three-day standard deviation on a \$1 billion position is

$$1{,}000 \times 0.15 \times \sqrt{3/252} = \$16.4\text{m}$$

and at three standard deviations the requirement is about \$49m.

Volatility then triples to 45%, not extreme by the standards of 2022 in energy or rates. The same calculation gives a three-day standard deviation of \$49.1m and a requirement of about \$147m.

The fund must find an additional \$98 million in cash, on a position it has not changed, within a day. If it holds the cash, nothing happens. If it does not, it must sell—and so must everyone else in the same position at the same moment.

Case Study 2.1 LME nickel, March 2022

On March 7 and 8, 2022 the price of nickel on the London Metal Exchange (LME) more than tripled: from about \$29,000 a metric ton at the previous Friday's close it rose some 66% on the first day and then, in a few hours of the second morning, spiked above \$100,000.

Behind the move was a very large short position held by a producer that had hedged its output. As prices rose, the margin calls on that position grew to a size that threatened both the holder and the brokers standing between it and the exchange. Nickel is not an unusually large market, and the losses were plausibly beyond what the default waterfall could have absorbed without severe consequences for members.

The LME's response was extraordinary: it suspended trading in nickel and *canceled*

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several billion dollars of trades that had already executed that morning, resetting the market to the previous evening's close. Trading did not resume normally for over a week.

The cancellation protected the clearing system at the direct expense of those who had bought during the spike, who had transacted in good faith on a functioning exchange and had their profits annulled. Litigation followed. Whatever the legal merits, the episode established something uncomfortable that the textbook description of futures markets omits: when the alternative is the failure of the clearing house, the finality of a trade is not absolute.

Two further episodes the same year made the same point without any cancellation of trades. European energy utilities that had sold futures against future output faced margin calls in the tens of billions of euros when gas and power prices rose several-fold, and UK pension funds running liability-driven investment met collateral calls by selling gilts, which drove yields higher and triggered further calls. In each case a hedge correct in every economic sense was defeated by the timing of the cash it demanded; Sections 8.9 and 8.10 work both episodes (Example 8.9 for energy, the liability-driven investment case study and Example 8.11 for the pension funds), and Sections 40.8 and 40.12 tell them.

What can be done. Regulators have pressed CCPs toward margin models that are less reactive: floors on the volatility estimate, buffers built in calm periods, and caps on how fast requirements may increase. Each reduces procyclicality at the cost of holding more margin permanently, which is a real cost borne in normal times to avoid a rare one. Where the balance should sit is unresolved, and reasonable people place it differently.

2.10 WHO ABSORBS A CCP'S LOSSES?

The waterfall of Section 2.8 is finite. If losses exhaust every layer, the CCP faces its own failure, and since clearing is now mandatory for large parts of the derivatives market, the failure of a major CCP would be a systemic event of the first order.

The tools available at the bottom of the waterfall are uncomfortable. **Assessment rights** allow the CCP to demand further contributions from surviving members, transmitting stress to firms that may already be struggling. **Variation margin gains haircutting** withholds part of the gains owed to members whose positions have profited, taking money from the winners to pay for the losers. **Tear-up** cancels contracts outright, leaving members who thought they were hedged suddenly unhedged.

Each of these is a way of allocating a loss that has already occurred. None creates resources. And each would be deployed in precisely the conditions in which it does most damage.

This has produced an unresolved policy debate with three broad positions. Some argue CCPs should hold much more of their own capital, so that they bear more of the consequence of setting margin too low. CCPs respond that their capital is small relative to the risks they manage and that heavier requirements would make clearing uneconomic. Others argue for an explicit public backstop, on the grounds that a systemically essential institution will be rescued in practice and that pretending otherwise mainly obscures who bears the risk. The objection is that an explicit backstop invites exactly the underpricing of risk it would then have to cover.

The question matters for this book because so much of what follows assumes cleared markets function. It is worth holding in mind that the assumption is a judgment about institutional resilience, not a mathematical fact.

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2.11 OTC MARKETS: CLEARED AND BILATERAL

Not everything trades on an exchange. The over-the-counter market, in which participants negotiate directly, remains larger than the exchange-traded market by notional and is where most of the instruments in this book are transacted.

Post-crisis reform pushed it toward the futures model along two tracks.

Standardized OTC derivatives must be centrally cleared. Interest rate swaps in the major currencies and index credit default swaps are cleared through CCPs that operate essentially as described above. This is why Chapter 7 can treat cleared swaps as the standard case. Where they are cleared has become a policy question as well: since June 2025 the European Union's revised European Market Infrastructure Regulation (EMIR) has required EU firms above the clearing threshold in euro- and zloty-denominated interest rate derivatives and euro short-term interest rate derivatives to hold an active account at a CCP in the EU. Under technical standards in force since February 2026, they must also clear at least five trades on average through it in each of their most-used contract types, a rule aimed at reducing the EU's reliance on clearing houses in London.

Non-cleared derivatives face mandatory margin. Trades that cannot be cleared (because they are bespoke) are subject to uncleared margin rules requiring both variation margin and, above certain thresholds, initial margin posted to a segregated third party. The intent is to remove the incentive to avoid clearing simply to avoid posting collateral.

The result is that the practical distinction between exchange-traded and OTC has narrowed considerably. What remains is the contract's flexibility: an OTC trade can be tailored to a hedging need exactly, while a futures contract cannot. That flexibility is why the OTC market persists, and the basis risk it eliminates is real, as Chapter 3 discusses.

And clearing has now been extended beyond derivatives altogether. In December 2023 the US Securities and Exchange Commission adopted rules requiring that eligible secondary-market transactions in US Treasury securities be cleared through a registered clearing agency: both outright purchases and sales, and the repurchase agreements that finance them. The requirement is being phased in, with the cash leg first and repo six months later: after a one-year extension in 2025, cash transactions must be cleared from December 31, 2026 and repo from June 30, 2027.

The reasoning belongs to this book rather than to securities-market infrastructure. A large share of Treasury repo was bilateral and uncleared, which meant leverage extended on privately negotiated haircuts, to positions no regulator could total. The cash–futures basis trade of Section 6.5 is financed that way, and March 2020 is what happens when the financing withdraws from a crowded one. The rule is a market-structure answer to a derivatives problem, which is why it appears in this chapter and not in a chapter on bonds.

The mechanism is the one already described. It is novation to a central counterparty, margin set by a published model rather than agreed across a telephone, and multilateral netting across members, applied to a market where the incumbent clearing agency, the Fixed Income Clearing Corporation, has long cleared a substantial share of interdealer activity while much client activity sat outside it. Access models matter as much as the mandate: a client can be sponsored into clearing by a member, and rules permitting execution with one dealer and clearing through another determine how much competition the change actually produces.

And the questions this chapter has already raised follow it across. Section 2.8's default waterfall and Section 2.9's procyclicality now apply to the financing market for the asset that prices every other dollar asset. Concentrating that risk in a clearing house does not remove it; it makes it measurable, mutualizes it among members, and raises the stakes on the margin model. Whether that is a good trade is the same question Section 2.10 asks, on a larger underlying.

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2.12 TRADERS, ORDERS, AND THE SHAPE OF THE MARKET

The three motives for trading, hedging, speculation, and arbitrage, are those of Section 1.9, which also makes the case that hedgers and speculators need each other. In a futures market the categories are less clean than they look (a dealer hedging inventory is simultaneously speculating on the basis), and what this section adds is how traders place their orders and what their trading produces.

Orders. A **market order** executes immediately at the best available price. A **limit order** executes only at a specified price or better, and may not execute at all; the trade-off between certainty of execution and certainty of price is the fundamental one, and no order type escapes it. A **stop order** becomes a market order once the price reaches a trigger, and is commonly used to cap losses, though in a fast market it executes at whatever price is available *after* the trigger, which may be far worse. A **stop-limit order** adds a limit to that price (a floor, for a stop to sell) at the risk of not executing at all, which in a falling market is precisely the outcome the stop was meant to prevent.

The nickel episode is a reminder that every order type assumes a functioning market, and inherits the fragility of that assumption.

Price discovery

Hedging is not the only function these markets serve, and arguably not the most important one. A liquid futures market with contracts extending years forward produces something no individual participant could: a continuously updated, publicly visible **forward curve** for the underlying: the settlement prices of the successive delivery months plotted against those months, so that a single line shows what the market will pay for the asset at every date it trades. The process by which trading assembles many participants' scattered information into such a price is **price discovery**. Chapter 5 works out what determines its shape, and Figure 5.2 draws the characteristic ones.

That curve is used by firms that never trade a single contract. An oil refiner deciding how much crude to process next quarter, a farmer choosing what to plant, and a utility deciding whether to contract for supply all read prices off a curve established by others' trading. The information is a public good produced as a by-product of private speculation, which is the strongest argument for tolerating speculation that serves no hedging purpose of its own.

Two cautions attach. A futures price is not a forecast of the future spot price; it is a price that clears today's market for future delivery, and Chapter 5 shows it equals the expected spot price only when the asset carries no systematic risk, differing from it otherwise by a risk premium. And the signal degrades exactly when it matters most: a limit-locked or suspended market produces no price at all, and the LME's nickel cancellation produced prices that were subsequently declared not to have happened.

2.13 FORWARDS VERSUS FUTURES

Table 2.3 summarizes the differences.

The economically significant difference is daily settlement: because gains and losses are reinvested or financed at the short rate prevailing on the day they arrive, a futures price can differ from the forward price on the same terms, as Section 5.12 shows and Chapter 30 quantifies through the convexity adjustment.

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Table 2.3: Forward and futures contracts compared.

Forward	Futures
Private contract between two parties	Traded on an exchange
Not standardized	Standardized
Usually one delivery date	Range of delivery dates
Settled at the end of the contract	Settled daily
Delivery or final cash settlement usual	Position usually closed before delivery
Some credit risk	Virtually no credit risk

SUMMARY

A futures contract standardizes everything except price, which permits a liquid secondary market in which positions are closed by offset rather than negotiation, and arbitrage forces convergence to spot at delivery. Volume counts contracts traded and open interest counts contracts outstanding, so the two series read together carry information a price series alone does not.

The exchange manages credit risk not by assessing it but by making it irrelevant: positions are marked to market daily and losses collected in cash before they accumulate, with a fall below the maintenance level triggering restoration to the initial level. Initial margin covers a high-confidence adverse move over the assumed close-out period and is computed on the portfolio, so a calendar spread can require a small fraction of what its two legs would need outright. That credit is what makes relative-value strategies viable at high leverage and what makes them fragile, since it is withdrawn precisely when the two legs stop moving together.

A central counterparty novates each trade into two, collapsing up to $n(n-1)/2$ bilateral exposures into n, permitting multilateral netting and an orderly default procedure. It concentrates risk rather than eliminating it. Access is two-tier, so most participants' protection depends on how their collateral is segregated and on whether they can be ported to a surviving member, which requires someone willing to take the book at the worst possible moment. Losses are absorbed in a fixed sequence on the principle that the defaulter pays first, and in the worked example the default reaches into the surviving members' pooled contributions: clearing distributed the loss rather than removing it. Beyond the waterfall, assessments, gain haircutting, and tear-ups allocate losses rather than creating resources.

Initial margin rises roughly in proportion to volatility, which is prudent individually and destabilizing collectively. The three episodes of 2022 (LME nickel, European energy, and UK liability-driven investment) shared that mechanism despite different causes: economically sound hedges were defeated by the timing of the cash they demanded. The nickel cancellation established that trade finality yields when the alternative is the clearing house failing. OTC markets have converged on this model, and what survives the convergence is flexibility: an OTC contract can match a hedging need exactly and a futures contract cannot.

FURTHER READING

The Committee on Payments and Market Infrastructures and the International Organization of Securities Commissions (CPMI–IOSCO) *Principles for Financial Market Infrastructures* (2012), together with the later guidance on CCP resilience and recovery, is the primary source for the clearing machinery of this chapter, and it is written for supervisors rather than for lawyers. Pirrong's ISDA discussion paper on the economics of central clearing is the skeptical counterweight to official-sector enthusiasm and is worth reading for exactly that reason. On the margin spiral of 2022, the BIS bulletin on margin dynamics in centrally cleared commodities markets and the Financial Stability Board's *Liquidity Preparedness for Margin and Collateral*

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Calls (2024) are the official-sector accounts, and Duffie's Jackson Hole paper on Treasury market resilience shows the same mechanism at work in the largest market there is.

PRACTICE QUESTIONS

2.1 Explain, using an arbitrage argument in both directions, why the futures price converges to the spot price at delivery.

2.2 An investor sells three silver futures contracts, each for 5,000 ounces, at $28.00 per ounce. Initial margin is $9,000 per contract and maintenance margin $7,000 per contract. Construct the margin account over five days if the settlement prices are $28.20, $28.55, $28.30, $28.95, and $28.70. On which days is there a margin call, for how much, and why is the account restored to the initial rather than the maintenance level?

2.3 Six firms trade with one another bilaterally. How many bilateral exposures exist, how many remain after novation to a CCP, and what is the general result for n firms?

2.4 A clearing member defaults, generating a close-out loss of $1.4 billion. The defaulter's margin is $600m, its default fund contribution $120m, the CCP's skin in the game $75m, and the surviving members' default fund $2.5 billion. Work through the waterfall in order, state what percentage of the surviving members' fund is consumed, and explain why the CCP's own capital sits ahead of surviving members' contributions.

2.5 A $500 million position has an initial margin requirement set at three standard deviations of a three-day move. Calculate the requirement at annualized volatilities of 12% and 30%, state how much additional cash must be found, and explain why a practice that is prudent for one clearing house may be destabilizing for the system.

2.6 Distinguish volume from open interest over three days: (i) two new participants open a 20-contract position between them; (ii) one of them closes 8 to a new entrant; (iii) both original parties close 12. Give both figures for each day.

2.7 A trader is long 200 September contracts and short 200 December, with outright margin of $1,500 per contract and a spread rate of $250 per pair. Compare the portfolio and contract-by-contract requirements, and explain what happens to the position if the spread relationship breaks down.

2.8 Explain the difference between individual and omnibus segregation of client collateral, why a client might not know which it has, and why the chance of being ported is lowest exactly when it is most needed.

2.9 Explain what a corner or squeeze is and how position limits are meant to prevent it. Why did the nickel episode produce similar market effects without being a manipulation, and what does the cancellation of executed trades imply about finality?

2.10 Futures markets are defended on the grounds that they produce price discovery. Explain what this means, who benefits, why it is an argument for tolerating speculation that serves no hedging purpose, and under what conditions the argument fails.

FURTHER QUESTIONS

2.11 The 2022 episodes in energy, nickel, and UK pensions had quite different causes. Identify the mechanism common to all three, explain why being correctly hedged did not protect the firms involved, and state what a risk manager should have measured that a mark-to-market framework does not capture.

2.12 Set out the case that CCPs should hold substantially more of their own capital in the waterfall, then the case against. What evidence would move you?

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2.13 Variation margin gains haircutting takes money owed to members whose positions have profited. Explain when it would be used, why it is preferred to simply failing, and what it does to the incentive to clear.

2.14 Central clearing concentrates exposures in a single institution. Construct the argument that this makes the system *more* fragile rather than less, and identify what would have to be true for the argument to be right.

2.15 A commodity producer hedges three years of output by selling futures. Prices triple. Show that the firm is economically better off than if it had not hedged, yet may fail. What financing arrangements would have prevented this, and what do they cost in normal times?

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